

For Loop Practice Set

1. Print numbers from 1 to n

➤ Write a program to print all numbers from 1 to a given number n using a for loop.

2. Multiplication Table

➤ Write a program to print the multiplication table of a given number up to 10.

3. Sum of first n natural numbers

➤ Write a program to calculate the sum of the first n natural numbers.

4. Print even numbers between 1 and n

➤ Write a program to print all even numbers between 1 and a given number n .

5. Print the square of first n numbers

➤ Write a program to print the squares of numbers from 1 to n .

6. Fibonacci Series up to n terms

➤ Write a program to print the first n terms of the Fibonacci series using a for loop.

7. LCM of two numbers

➤ Write a program to find the Least Common Multiple (LCM) of two given numbers using a for loop.

8. HCF (GCD) of two numbers

➤ Write a program to calculate the Highest Common Factor (HCF) or Greatest Common Divisor (GCD) of two given numbers using a for loop.

9. Check if a number is Prime

➤ Write a program to check whether a given number is prime using a for loop.

10. Count the number of digits in a number

➤ Write a program to count how many digits a given integer contains.

11. Reverse a number

➤ Write a program to reverse a given number using a for loop.

12. Print all Prime numbers between 1 and n

➤ Write a program to print all prime numbers between 1 and a given number n using nested for loops.

13. Armstrong numbers in a range

➤ Write a program to print all Armstrong numbers between 1 and n .

14. Factorial of a number

➤ Write a program to calculate the factorial of a given number using a for loop.

15. Check if a number is a palindrome

➤ Write a program to check whether a number is a palindrome using a for loop.

16. Find the sum of digits of a number

➤ Write a program to calculate the sum of all digits in a given number.

Print the following Patterns:

1. Print a Single Star (*)

2. Print Five Stars (*****)

3. Print n Stars on Same Line

4. Print Square of Stars ($n \times n$ Stars)

```
*****
*****
*****
*****
*****
```

5. Print an Increasing Triangle of Stars

```
 *
**
***
****
*****
```

6. Print a Right-Aligned Triangle of Stars

```
  *  
 **  
 ***  
 ****  
 *****
```

7. Print Stars in Even Numbers (2, 4, 6, 8, 10)

```
**  
****  
*****  
*****  
*****
```

8. Print Stars in Odd Numbers (1, 3, 5, 7, 9)

```
*  
***  
*****  
*****  
*****
```

9. Print a Centered Pyramid of Stars

```
  *  
 ***  
 *****  
 *****  
 *****
```

11. Print Numbers in an Increasing Sequence (1, 12, 123, 1234, 12345)

```
1
12
123
1234
12345
```

12. Print Repeated Numbers per Row (Same Number Repeated)

```
1
22
333
4444
55555
```

13. Print the following pattern.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

14. Print the following pattern.

```
1
2 3
4 5 6
7 8 9 0
1 2 3 4 5
5 7 8 9 0 1
2 3 4 5 6 7 8
```

15. Print the following pattern.

```

1
0 1
0 1 0
1 0 1 0
1 0 1 0 1

```

16. Print the following pattern.

```

A
B C
D E F
G H I J
K L M N O

```

17. Print the following pattern.

```

A
B B
C C C
D D D D
E E E E E

```

18. Print the following pattern.

```

*
**
****
*****
****
***
**
*

```

19. Print the following pattern.



20. Print the following pattern.

